P8X Backplane Bus Definition

Revision C — June 2026 — FOR REVIEW

Connector: DIN 41612, 3 rows (A/B/C) × 32 pins. Backplane: 10 slots, 25.4 mm pitch, 4-layer (signals / GND plane / +5V plane / signals). Power: A1,B1,C1,A2,B2,C2 = +5V; A31,B31,C31,A32,B32,C32 = GND. Row B pins 3–30 are a grounded guard row between the two signal rows. This table is generated from the same pin-map function used by the CAD generators, so it cannot drift from the board files.

Pin Assignment (component-side view of card connector)

Pin	Row A	Row B	Row C
1	+5V	+5V	+5V
2	+5V	+5V	+5V
3	D0	GND	A0
4	D1	GND	A1
5	D2	GND	A2
6	D3	GND	A3
7	D4	GND	A4
8	D5	GND	A5
9	D6	GND	A6
10	D7	GND	A7
11	-RES	GND	A8
12	DOE0	GND	A9
13	DOE1	GND	A10
14	DOE2	GND	A11
15	DOE3	GND	A12
16	DLD0	GND	A13
17	DLD1	GND	A14
18	DLD2	GND	A15
19	DLD3	GND	ALUS0
20	PSEL0	GND	ALUS1
21	PSEL1	GND	ALUS2
22	PINC	GND	ALUS3
23	PDEC	GND	ALUM
24	CLK	GND	CIN
25	CLKB	GND	SH0
26	LDF	GND	SH1
27	FC	SPARE8	SPARE4
28	FZ	SPARE9	SPARE5
29	FN	SPARE10	SPARE6
30	FV	SPARE11	SPARE7
31	GND	GND	GND
32	GND	GND	GND

Shading: red = +5V, blue = GND, grey = spare (bused).

Legend

ning	Acronym	Meaning		
bus, bit n (8-bit)	An	Address bus, bit n (16-bit)		
Output Enable field: selects n card drives the data bus	DLD0-3	Data LoaD field: selects which register latches the data bus		
er SELect: which of P0-P3 ptr/ptr/SP) is active	PINC / PDEC	Pointer INCrement / DECrement (applies to selected pointer)		
1 ALU function Select lines	ALUM	74181 Mode: logic vs arithmetic		
Carry IN	SH0-1	SHifter control (pass/left/right/rotate)		
Flags: latch C,Z,N,V from ALU	-RES	RESet, active low (_N = active low)		
m CLocK and its complement -Bar); loads on rising CLK, strobes gated by CLKB	SPAREn	Unassigned, bused to all 10 slots, reserved for future use	FC FZ FN FV	ALU flags to control (condition mux), allo former SPARE0-3

Signal Dictionary

Signal	Pin(s)	Dir*	Description
D0	A3	<>	Data bus bit 0
D1	A4	<>	Data bus bit 1
D2	A5	<>	Data bus bit 2
D3	A6	<>	Data bus bit 3
D4	A7	<>	Data bus bit 4
D5	A8	<>	Data bus bit 5
D6	A9	<>	Data bus bit 6
D7	A10	<>	Data bus bit 7
A0	C3	RB>	Address bus bit 0
A1	C4	RB>	Address bus bit 1
A2	C5	RB>	Address bus bit 2
A3	C6	RB>	Address bus bit 3
A4	C7	RB>	Address bus bit 4
A5	C8	RB>	Address bus bit 5
A6	C9	RB>	Address bus bit 6
A7	C10	RB>	Address bus bit 7
A8	C11	RB>	Address bus bit 8
A9	C12	RB>	Address bus bit 9
A10	C13	RB>	Address bus bit 10
A11	C14	RB>	Address bus bit 11
A12	C15	RB>	Address bus bit 12
A13	C16	RB>	Address bus bit 13
A14	C17	RB>	Address bus bit 14
A15	C18	RB>	Address bus bit 15
DOE0	A12	<>	Data bus bit OE0
DOE1	A13	<>	Data bus bit OE1
DOE2	A14	<>	Data bus bit OE2
DOE3	A15	<>	Data bus bit OE3
DLD0	A16	<>	Data bus bit LD0
DLD1	A17	<>	Data bus bit LD1
DLD2	A18	<>	Data bus bit LD2
DLD3	A19	<>	Data bus bit LD3
PSEL0	A20	C>	Pointer select bit 0
PSEL1	A21	C>	Pointer select bit 1
PINC	A22	C>	Selected pointer increment
PDEC	A23	C>	Selected pointer decrement
LDF	A26	C>	Latch flags (ALU card)
ALUS0	C19	C>	74181 function select S0
ALUS1	C20	C>	74181 function select S1
ALUS2	C21	C>	74181 function select S2
ALUS3	C22	C>	74181 function select S3

Signal	Pin(s)	Dir*	Description
ALUM	C23	C>	74181 mode (logic/arith)
CIN	C24	C>	ALU carry in
SH0	C25	C>	Shifter control 0
SH1	C26	C>	Shifter control 1
CLK	A24	C>	System clock
CLKB	A25	C>	Inverted clock
-RES	A11	C>	System reset, active low
FC	A27	ALU>	Flag: Carry. Driven continuously by the ALU card flag register; read by the control car
FZ	A28	ALU>	Flag: Zero. ALU card to control card, as FC
FN	A29	ALU>	Flag: Negative (result bit 7). ALU card to control card, as FC
FV	A30	ALU>	Flag: oVerflow. ALU card to control card; rev A ALU card drives 0 (V unimplemented,
SPARE4	C27	n/a	Reserved, bused across all slots
SPARE5	C28	n/a	Reserved, bused across all slots
SPARE6	C29	n/a	Reserved, bused across all slots
SPARE7	C30	n/a	Reserved, bused across all slots
SPARE8	B27	n/a	Reserved, bused across all slots
SPARE9	B28	n/a	Reserved, bused across all slots
SPARE10	B29	n/a	Reserved, bused across all slots
SPARE11	B30	n/a	Reserved, bused across all slots
+5V	A1, A2, B1, B2, C1, C2	PWR	Power, 6 pins total
GND	A31 ... C32 (30 pins)	PWR	Ground / row-B guard

Dir Field Legend

Symbol	Meaning
C>	Driven by the Control card; received by all other cards
RB>	Driven by the Register Bank card (sole address-bus driver)
ALU>	Driven by the ALU card (flag register outputs)
<>	Bidirectional: exactly one driver per microcycle, enforced by one-hot DOE decode
PWR	Power distribution pin (+5V or GND planes)
n/a	Spare: bused across all slots, no driver assigned

DOE / DLD Field Encodings

Code	DOE: bus source	DLD: bus destination
0	(bus idle - pulled up on backplane)	none
1	A register	A register
2	B register	B register
3	T (hidden temp)	T (hidden temp)
4	T2 (hidden temp)	T2 (hidden temp)
5	ALU result via shifter	FLAGS (restore)
6	FLAGS	IR (instruction register)
7	MEM read (memory / I/O)	MEMW write strobe (memory / I/O)
8	PTRL - selected pointer low byte	PTRL - load selected pointer low
9	PTRH - selected pointer high byte	PTRH - load selected pointer high
10-15	reserved	reserved

Review Notes

1. RC clock termination (backplane far end) ships DNP; populate only if scope shows ringing.
2. D0-D7 carry 10k pull-ups on the backplane only; cards never add bus conditioning.
3. Loads occur on rising CLK; write strobes are gated with CLKB (second half-cycle).
4. Address bus is driven exclusively by the register-bank card (selected pointer).
5. SPARE4-7 (row C) and SPARE8-11 (row B27-B30) are bused across all slots, reserved for future allocation.

5b. SPARE0-3 were reallocated as flag lines FC/FZ/FN/FV (A27-A30). SPARE numbering therefore starts at 4.

5c. Row B ground guard now spans B3-B26; B27-B30 carry SPARE8-11.

6. Verify row A/C orientation against physical DIN connectors before first fab.